Major Architecture Refactor: Move Assessment Experience to Dedicated Report Page

IMPORTANT

This is a structural refactor of the application's primary user journey.

Do NOT start coding immediately.

First understand the desired architecture, then implement incrementally.

The existing calculator logic, appliance data, recommendation engines, health score engines, PDF generation, and insight pages already work correctly.

The goal is NOT to rebuild functionality.

The goal is to improve UX, scalability, SEO, maintainability, and desktop experience.

Current Problem

The calculator page currently contains:

- Appliance selection
- Load summary
- Smart insights
- Health score
- Consumption breakdown
- Alerts
- Savings opportunities
- Energy cost estimator

All displayed simultaneously.

This causes:

- Overloaded desktop sidebar
- Long sticky panels
- Reduced readability
- Difficult future expansion
- Limited SEO value because recommendations are hidden inside the calculator page

The application is beginning to behave like an Electrical Consultant Platform rather than a simple calculator.

The UI architecture must reflect this evolution.

New Product Architecture

The application should be divided into three layers.

Layer 1

Calculator

↓

Layer 2

Assessment Report

↓

Layer 3

Recommendation Detail Pages

Desired User Flow

User lands on:

/

Electrical Load Calculator

↓

Select appliances

↓

Generate Assessment

↓

Navigate to:

/assessment

↓

Review complete assessment

↓

Open recommendation pages

↓

/recommendations/mcb

/recommendations/cable

/recommendations/inverter

/recommendations/solar

Core Rule

Calculator Page

= Input Experience

Assessment Page

= Analysis Experience

Recommendation Pages

= Educational + Explanation Experience

Never mix all three responsibilities into one page.

STEP 1

Refactor Calculator Page

Current calculator page contains:

- Appliance inputs
- Assessment cards
- Insights
- Health score
- Charts
- Alerts
- Cost estimator

This must change.

Calculator Page Responsibilities

Keep:

- Appliance selection
- Quick load summary
- Quick consumption summary
- Top consumer
- Health score snapshot

Only display the most important metrics.

Do NOT show:

- Smart insights cards
- Consumption breakdown charts
- Alerts
- Savings recommendations
- Energy cost estimator
- Detailed analysis

Remove these from calculator page.

New Desktop Layout

Calculator Page

LEFT SIDE

Appliance Selection

RIGHT SIDE (Sticky Summary)

Load

Consumption

Top Consumer

Health Score

Generate Assessment Button

Nothing more.

Keep the page focused.

Generate Assessment CTA

Replace current report button with:

Generate Full Assessment

or

View Full Assessment

This becomes the primary CTA.

STEP 2

Create Dedicated Assessment Page

Route:

/assessment

This becomes the most important page in the application.

Think of it as:

Electrical Assessment Report

not

Results Page

Assessment Page Layout

Section 1

Assessment Header

Display:

Electrical Assessment Report

Generated from your appliance configuration

Section 2

Load Summary

Display:

Total Load

Daily Consumption

Monthly Consumption

Top Consumer

Health Score

Section 3

Smart Recommendations

Display cards:

Recommended MCB

Recommended Cable

Recommended Inverter

Recommended Solar

Each card contains:

Recommendation

Short explanation

View Details button

Section 4

Consumption Analysis

Display:

Category breakdown

Charts

Distribution

Highest consuming category

Section 5

Electrical Health

Display:

Health Score

Scoring breakdown

Risk indicators

Recommendations

Section 6

High Consumption Devices

Display flagged appliances.

Section 7

Energy Saving Opportunities

Display savings recommendations.

Section 8

Energy Cost Estimator

Optional feature.

Only estimate cost if user provides rate.

Do not assume tariffs.

Section 9

Download Report

Generate PDF

Assessment Page Design Requirements

Desktop:

Use full-width content.

Do NOT use narrow sidebar cards.

Assessment should feel like a professional audit report.

Maximum content width:

1200–1400px

Use sections with generous spacing.

STEP 3

Recommendation Pages

Create recommendation hub structure.

Routes:

/recommendations/mcb

/recommendations/cable

/recommendations/inverter

/recommendations/solar

IMPORTANT

Do not break existing recommendation pages.

Refactor them to receive assessment data.

Data Passing Strategy

Preferred order:

1. Global state
2. Context
3. URL-safe persisted storage

Avoid:

Massive query strings.

Avoid recalculating recommendations unnecessarily.

Recommendations should use the same data generated during assessment.

Single source of truth.

STEP 4

Breadcrumb Architecture

Assessment Page

Home > Assessment

MCB Page

Home > Assessment > MCB Recommendation

Cable Page

Home > Assessment > Cable Recommendation

Inverter Page

Home > Assessment > Inverter Recommendation

Solar Page

Home > Assessment > Solar Recommendation

Use semantic breadcrumb markup.

STEP 5

Assessment as Central Hub

Assessment page becomes the hub.

All recommendation pages should originate from Assessment.

Assessment links to:

MCB

Cable

Inverter

Solar

Every recommendation page should link back to:

Assessment

and also show:

Related Recommendations

Example:

On MCB page:

Related:

Cable Recommendation

Solar Recommendation

Inverter Recommendation

STEP 6

Routing Safety Requirements

This refactor is high risk.

Protect existing functionality.

Requirements:

Do not break calculator logic.

Do not break appliance state.

Do not break PDF generation.

Do not break recommendation engines.

Do not break health score calculations.

Do not break energy estimator.

Do not duplicate business logic.

STEP 7

Migration Strategy

Implement incrementally.

Phase A

Create assessment route.

Phase B

Move report components.

Phase C

Connect navigation.

Phase D

Refactor recommendation navigation.

Phase E

Remove duplicated components from calculator page.

Do NOT remove existing functionality until replacement is verified working.

STEP 8

SEO Requirements

Assessment Page

Target:

Electrical Assessment Report

Electrical Load Assessment

Electrical Consumption Analysis

Electrical Planning Report

Recommendation Pages

Target:

MCB Recommendation

Cable Recommendation

Inverter Recommendation

Solar Recommendation

Add:

Breadcrumb schema

FAQ schema

SoftwareApplication schema

Internal linking

Canonical URLs

Success Criteria

The final experience should feel like:

Step 1

Build your electrical profile

↓

Step 2

Receive a professional electrical assessment

↓

Step 3

Understand each recommendation in detail

The calculator page should become cleaner.

The assessment page should become richer.

The recommendation pages should become more educational.

The entire application should feel like an Electrical Consultant Platform rather than a collection of widgets and cards.